A Hamiltonian–Geometric Optimization Framework for Machine Learning

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Abstract

This report consolidates three things about the Hamiltonian–geometric optimizer proposed in *"A Hamiltonian–Geometric Optimization Framework for Machine Learning"*: (1) the mathematical background of the framework; (2) an analytical validation of the update equations against the underlying Hamiltonian mechanics; and (3) experimental results obtained by running the proposed optimizer, and its competitors, on multiple local benchmarks, plus a direct visualization of the symmetry structure the framework depends on. A dedicated derivation shows how gradient descent, momentum, Adam/RMSProp-style diagonal preconditioning, natural-gradient-style methods, entropy descent, and the full Hamiltonian-geometric optimizer sit inside one update family; it also gives Adam's diagonal-metric update from a local variational step principle, and a further derivation extends the same update family to modern optimizers (AdamW, Lion, Shampoo/K-FAC, and Muon), each obtained by a specific choice of metric, kinetic term, or potential splitting rather than by analogy. The analysis also fixes the sign convention for the spectral Hamiltonian and supplies an explicit vector-to-matrix construction for the memory metric M_{ij}.

1 Mathematical Background

1.1 Coordinates, potential, and the Hamiltonian

The framework treats model parameters $\theta = (\theta^1, \dots, \theta^n)$ as generalized coordinates $q^i \equiv \theta^i$, with the loss function playing the role of potential energy, $V(q) \equiv L(\theta)$. Given a metric $g_{ij}(\theta)$ on parameter space defining a kinetic term

$$T = \tfrac{1}{2}\,\dot\theta^i\, g_{ij}(\theta)\,\dot\theta^j, \tag{1}$$

the Legendre transform of the Lagrangian $\mathcal{L} = T - V$ gives canonical momentum $p_i = \partial\mathcal{L}/\partial\dot\theta^i = g_{ij}\dot\theta^j$ and Hamiltonian

$$H(\theta, p) = \tfrac{1}{2}\,p_i\, g^{ij}(\theta)\,p_j + L(\theta), \tag{2}$$

where g^{ij} is the matrix inverse of g_{ij}, $g^{ij}g_{jk} = \delta^i_k$.

1.2 Hamilton's equations and the geometric force

Hamilton's equations $\dot\theta^i = \partial H/\partial p_i$, $\dot p_i = -\partial H/\partial\theta^i$ applied to Eq. (2) give

$$\dot\theta^i = g^{ij}(\theta)\,p_j, \tag{3}$$
$$\dot p_i = -\partial_i L - \tfrac{1}{2}\,p_j p_k\,\partial_i g^{jk}. \tag{4}$$

The second term in Eq. (4) is named the *geometric force*,

$$F_{\text{geo}} \;=\; \tfrac{1}{2}\,p^{\mathsf T}\big(\nabla_\theta g^{-1}\big)\,p, \tag{5}$$

so that Eq. (4) reads $\dot p_i = -\partial_i L - F_{\text{geo}}$: motion is driven by the loss gradient *and* by how the metric itself curves across parameter space.

1.3 Dissipation

Machine-learning optimization is not conservative – the loss must decrease – so a Rayleigh dissipation function $\mathcal{R}(p) = \frac{1}{2}\gamma\,p^i p_i$ is added, giving $\dot p_i = -\partial H/\partial\theta^i - \partial\mathcal{R}/\partial p_i$, i.e.

$$\dot\theta^i = g^{ij}(\theta)\,p_j, \qquad \dot p_i = -\partial_i L - F_{\text{geo}} - \gamma p_i. \tag{6}$$

In the flat Euclidean case ($g_{ij} = \delta_{ij}$, so $\partial_i g^{jk} = 0$ identically and $F_{\text{geo}} = 0$), Eq. (6) reduces to $\dot\theta = p$, $\dot p = -\nabla_\theta L - \gamma p$, and in the overdamped limit ($p \approx -\gamma^{-1}\nabla_\theta L$) this Euler-discretizes to plain gradient descent. Classical momentum and Adam are presented as further special cases: momentum as damped Hamiltonian flow with $p_t \sim v_t$; Adam as an approximate *diagonal*-metric Hamiltonian optimizer, $g_t^{ij} \approx \delta^{ij}/(\sqrt{v_t^i} + \epsilon)$, with the explicit caveat that Adam has no analogue of F_{geo} since v_t is treated as locally constant.

1.4 The generalized update rule

With a memory (non-Markovian) correction built from an exponentially weighted kernel over past gradients,

$$F_{\text{mem}}(\theta_t) = \mu\sum_{k=1}^{t}\kappa^{\,t-k}\,\nabla_\theta L(\theta_k), \qquad 0 < \kappa < 1, \tag{7}$$

the proposed discretized optimizer is

$$p_{t+1} = \beta p_t - \eta\Big[\nabla_\theta L + F_{\text{geo}} + F_{\text{mem}} + \alpha\nabla_\theta S(g)\Big], \tag{8}$$
$$\theta_{t+1} = \theta_t + \eta\,g^{-1}\,p_{t+1}. \tag{9}$$

The metric itself may be built from the (regularized) Hessian of the loss, $g_{ij} = H_{ij} + \lambda\delta_{ij}$, $H_{ij} = \partial^2 L/\partial\theta^i\partial\theta^j$, $\lambda > 0$.

Completing the memory-metric term M_{ij}

The source paper additionally proposes $g_{ij} = H_{ij} + \mu M_{ij}$, calling M_{ij} "the memory term implied by" Eq. (7), but Eq. (7) defines F_{mem} as a *vector* (one free index, shape-compatible with $\nabla_\theta L$), so it cannot be M_{ij} itself – a rank-2, symmetric object is needed to add to H_{ij}. Use the same exponentially weighted construction Eq. (7) already uses, applied one power higher, to the *outer product* of the gradient with itself rather than to the gradient alone:

$$M_{ij}(\theta_t) = \kappa\,M_{ij}(\theta_{t-1}) + \big(\partial_i L(\theta_t)\big)\big(\partial_j L(\theta_t)\big), \tag{10}$$

Table 1: Optimizer families as limits or approximations of the Hamiltonian–geometric update.

Optimizer family	Hamiltonian specialization	What is removed or approximated
Gradient descent	Flat metric, overdamped limit	Momentum relaxes instantly; $F_{\text{geo}} = 0$
Heavy-ball momentum	Flat metric with finite momentum	No curvature, memory, or spectral force
Nesterov-style momentum	Flat damped flow with lookahead gradient	Lookahead discretization of momentum flow
RMSProp / Adam	Diagonal adaptive metric approximation	Ignores F_{geo} from metric variation
Natural gradient	Full information metric	Usually no Hamiltonian momentum or memory force
Entropy descent	Full/preconditioned metric step	Keeps g^{-1} but drops momentum and memory
Hamiltonian-geometric	Full metric with optional forces	Retains momentum, metric, memory, entropy, curvature

with the same decay κ as Eq. (7) (so no new hyperparameter is introduced) and $M_{ij}(\theta_0) = 0$. This is exactly the uncentered-empirical-Fisher construction the paper already invokes informally to motivate Adam's diagonal metric below (§2, "RMSProp and Adam") ($v_t \approx$ diag of an EMA of squared gradients) – Eq. (10) is the same recursion kept as a full matrix instead of truncated to its diagonal. Three properties make it a legitimate metric contribution rather than an arbitrary patch: (1) it is manifestly symmetric, $M_{ij} = M_{ji}$, since $(\partial_i L)(\partial_j L) = (\partial_j L)(\partial_i L)$; (2) it is positive semidefinite for any gradient sequence, since it is a nonnegative sum of rank-1 PSD terms vv^T, so $g_{ij} = H_{ij} + \lambda\delta_{ij} + \mu M_{ij}$ can only add curvature, never break the positive-definiteness already established for H; and (3) it reduces to exactly F_{mem}'s own recursion in the one-dimensional case, $M(\theta_t) = \kappa M(\theta_{t-1}) + (\partial L)^2$, matching Eq. (7) squared term-by-term. This construction is implemented (opt-in, off by default) as `HamiltonianGeometricConfig`'s `use_memory_metric`/`memory_metric_coupling` and verified in `tests/test_hamiltonian_geometric.py` to stay symmetric PSD and to match the closed-form recursion exactly (see §4).

2 Standard Optimizers as Special Cases

The unifying claim of the source paper is not merely philosophical: the usual optimizer families appear by choosing particular metrics, damping limits, and force terms in the Hamiltonian–geometric update. The master form is

$$p_{t+1} = \beta p_t - \eta \left[\nabla_\theta L + F_{\text{geo}} + F_{\text{mem}} + \alpha \nabla_\theta S(g)\right], \quad (11)$$

$$\theta_{t+1} = \theta_t + \eta g^{-1}(\theta_t) p_{t+1}. \quad (12)$$

Different optimizers are obtained by deleting or restricting pieces of this system:

Gradient descent

Set the metric to the Euclidean metric, $g = I$, remove memory and spectral forces, and note that $\nabla_\theta g^{-1} = 0$, so $F_{\text{geo}} = 0$. The dissipative continuous-time Hamiltonian system reduces to

$$\dot\theta = p, \qquad \dot p = -\nabla_\theta L - \gamma p. \quad (13)$$

In the overdamped limit, momentum equilibrates much faster than position: $\dot p \approx 0$, hence

$$p \approx -\gamma^{-1} \nabla_\theta L. \quad (14)$$

Substitution into $\dot\theta = p$ gives

$$\dot\theta \approx -\gamma^{-1} \nabla_\theta L. \quad (15)$$

Forward Euler with step size h therefore gives

$$\theta_{t+1} = \theta_t - \frac{h}{\gamma} \nabla_\theta L(\theta_t) = \theta_t - \eta \nabla_\theta L(\theta_t), \quad (16)$$

which is ordinary gradient descent.

Heavy-ball momentum

Keep the same flat metric $g = I$ and $F_{\text{geo}} = F_{\text{mem}} = \alpha \nabla S(g) = 0$, but do not take the overdamped limit. The master update becomes

$$p_{t+1} = \beta p_t - \eta \nabla_\theta L(\theta_t), \quad (17)$$

$$\theta_{t+1} = \theta_t + \eta p_{t+1}. \quad (18)$$

Define the usual descent velocity $v_t = -p_t$. Then

$$v_{t+1} = \beta v_t + \eta \nabla_\theta L(\theta_t), \quad (19)$$

$$\theta_{t+1} = \theta_t - \eta v_{t+1}. \quad (20)$$

Up to the conventional placement of the learning rate inside or outside the velocity variable, this is the heavy-ball momentum update.

Nesterov-style momentum

Nesterov momentum uses the same flat damped Hamiltonian skeleton but evaluates the gradient at a lookahead point. Starting from the heavy-ball reduction, replace the force evaluation point by

$$\tilde\theta_t = \theta_t - \beta v_t. \quad (21)$$

The update becomes

$$v_{t+1} = \beta v_t + \eta \nabla_\theta L(\tilde\theta_t), \quad (22)$$

$$\theta_{t+1} = \theta_t - \eta v_{t+1}. \quad (23)$$

Thus Nesterov is not a new metric; it is a lookahead discretization of the same flat dissipative Hamiltonian momentum flow.

RMSProp and Adam

Let the metric be diagonal and time-adaptive:

$$g_t = \mathrm{diag}\left(\sqrt{\hat{v}_t} + \epsilon\right), \qquad g_t^{-1} = \mathrm{diag}\left((\sqrt{\hat{v}_t} + \epsilon)^{-1}\right), \tag{24}$$

where $\hat{v}_t$ is an exponential moving average of squared gradients. With momentum m_t identified with the descent momentum, the metric-preconditioned step is

$$\theta_{t+1} = \theta_t - \eta\, g_t^{-1} m_t = \theta_t - \eta\, \frac{m_t}{\sqrt{\hat{v}_t} + \epsilon}. \tag{25}$$

This is the Adam/RMSProp form. The correspondence is deliberately qualified: Adam treats g_t as locally fixed during the step, so it drops

$$F_{\mathrm{geo}} = \tfrac{1}{2} p^{\mathsf{T}} (\nabla_\theta g^{-1}) p. \tag{26}$$

Therefore Adam is a diagonal-metric approximation to the Hamiltonian optimizer, not the full Hamiltonian-geometric optimizer.

The same Adam form can also be derived variationally from a local metric-weighted step principle. At iteration t, approximate the loss by its first-order expansion around θ_t and penalize motion using a positive diagonal metric G_t:

$$\Delta\theta_t = \arg\min_{\Delta} \left[m_t^{\mathsf{T}} \Delta + \frac{1}{2\eta} \Delta^{\mathsf{T}} G_t \Delta \right]. \tag{27}$$

Here m_t is the exponentially averaged gradient direction and G_t is the local geometry used to measure the cost of a parameter displacement. The stationarity condition is

$$0 = \nabla_\Delta \left[m_t^{\mathsf{T}} \Delta + \frac{1}{2\eta} \Delta^{\mathsf{T}} G_t \Delta \right] = m_t + \eta^{-1} G_t \Delta, \tag{28}$$

so

$$\Delta\theta_t = -\eta G_t^{-1} m_t, \qquad \theta_{t+1} = \theta_t + \Delta\theta_t. \tag{29}$$

Choosing

$$G_t = \mathrm{diag}\left(\sqrt{\hat{v}_t} + \epsilon\right) \tag{30}$$

gives

$$\theta_{t+1} = \theta_t - \eta\, \frac{m_t}{\sqrt{\hat{v}_t} + \epsilon}, \tag{31}$$

which is Adam's parameter update after bias-correcting m_t and $\hat{v}_t$. This derivation makes the variational meaning explicit: Adam chooses the step that most decreases the local linearized loss per unit diagonal-metric movement cost. In Hamiltonian language, it is the diagonal, locally frozen, metric-preconditioned variational step. What it still does not include is the curvature force that appears when the metric is differentiated as a function of the parameters.

Natural gradient

Choose the metric to be an information metric, usually the Fisher information matrix $F(\theta)$, and remove Hamiltonian momentum, memory, spectral force, and geometric-force dynamics. The pure metric-preconditioned descent step is

$$\theta_{t+1} = \theta_t - \eta\, F(\theta_t)^{-1} \nabla_\theta L(\theta_t), \tag{32}$$

which is natural gradient descent. It is the full-metric, no-momentum limit of the same geometry.

Entropy descent

Use a positive-definite metric $g(\theta)$ and keep the inverse-metric preconditioner and spectral diagnostic, but set $\beta = 0$ and remove memory and the explicit geometric force. Then

$$p_{t+1} = -\eta \nabla_\theta L(\theta_t), \qquad \theta_{t+1} = \theta_t - \eta^2 g^{-1}(\theta_t) \nabla_\theta L(\theta_t). \tag{33}$$

Absorbing one factor of η into the effective step size gives

$$\theta_{t+1} = \theta_t - \eta_{\mathrm{eff}}\, g^{-1}(\theta_t) \nabla_\theta L(\theta_t), \tag{34}$$

the entropy-descent / metric-preconditioned update used in the experiments.

The Hamiltonian-geometric optimizer is the general member of this family in the present implementation: it contains the metric, momentum, and optional curvature, memory, and spectral forces in one update law.

3 Modern Optimizers as Further Derivations

§2 covers the optimizer families the source paper itself discusses (gradient descent, momentum, Adam/RMSProp, natural gradient, entropy descent), all published before or around the paper's own reference points. Whether the framework is a genuine *unification* rather than a retrofit onto optimizers it was designed to reproduce is better tested against methods published independently of it, with no shared authorship or design intent. Four such optimizers are derived below, each as a specific, motivated choice of metric, kinetic term, or potential splitting – not by loose analogy – plus two more (AdaFactor, LAMB/LARS) that complete the picture of how much metric structure each family keeps.

AdamW: decoupled weight decay as a split metric

AdamW (Loshchilov & Hutter, *Decoupled Weight Decay Regularization*, ICLR 2019) updates

$$\theta_{t+1} = \theta_t - \eta \left[\frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon} + \lambda\theta_t \right], \tag{35}$$

and is empirically distinct from plain Adam applied to the L_2-regularized loss $L(\theta) + \frac{\lambda}{2}\|\theta\|^2$, whose gradient $\nabla_\theta L + \lambda\theta$ gets divided by $\sqrt{\hat{v}_t} + \epsilon$ *in its entirety* – the regularizer term is contaminated by the adaptive scaling meant for the loss term. Eq. (35) keeps the two terms separately scaled. This is exactly the potential splitting a Hamiltonian system uses for a separable potential $V = V_1 + V_2$: evolve each piece with its *own* metric via Lie–Trotter operator splitting rather than sharing one metric across the whole potential. Take $V_1 = L(\theta)$ with the diagonal Adam metric $g_1^{-1} = \mathrm{diag}((\sqrt{\hat{v}_t} + \epsilon)^{-1})$ from §2, and $V_2 = \frac{\lambda}{2}\|\theta\|^2$ with the *flat* metric $g_2 = I$ (the same flat-metric case that reduces the framework to plain gradient descent, §2). One Lie–Trotter half-step under each potential,

$$\theta' = \theta_t - \eta\, g_1^{-1} \nabla_\theta L(\theta_t), \qquad \theta_{t+1} = \theta' - \eta\, g_2^{-1}(\lambda\theta'), \tag{36}$$

reproduces Eq. (35) to first order in η ($\theta' \approx \theta_t$ inside the second half-step). AdamW is therefore the framework with a *piecewise* metric – adaptive on the data-fit term, flat on the regularizer – rather than a single shared metric, which is precisely the mechanism the ML literature already uses informally to explain why decoupled weight decay differs from L_2 regularization in adaptive methods, now expressed as a metric choice.

Lion: an L_1 kinetic term

Lion (Chen et al., *Symbolic Discovery of Optimization Algorithms*, NeurIPS 2023, arXiv:2302.06675) updates

$$\begin{aligned} c_t &= \beta_1 m_{t-1} + (1-\beta_1)\nabla_\theta L(\theta_t), \\ \theta_{t+1} &= \theta_t - \eta\,\mathrm{sign}(c_t), \\ m_t &= \beta_2 m_{t-1} + (1-\beta_2)\nabla_\theta L(\theta_t). \end{aligned} \tag{37}$$

Every optimizer in §2 keeps the Hamiltonian's kinetic term quadratic, $T = \frac{1}{2}\dot\theta^i g_{ij}\dot\theta^j$, so $\dot\theta = \partial H/\partial p = g^{-1}p$ is always linear in p. Lion's sign$(\cdot)$ dynamics require dropping that assumption: take the flat metric $g = I$ (so $F_{\text{geo}} = 0$, as in gradient descent) but replace the quadratic kinetic term with the L_1 kinetic term $T = \sum_i |p_i|$. The Legendre transform still applies to a non-quadratic convex kinetic term; Hamilton's equation for θ becomes

$$\dot\theta^i = \frac{\partial T}{\partial p_i} = \mathrm{sign}(p_i), \tag{38}$$

the subgradient of $|p_i|$ at $p_i \neq 0$, while $\dot p = -\nabla_\theta L$ is unchanged (Eq. (6)'s flat-metric case, without the dissipation term since Lion has no separate β damping coefficient distinct from its momentum decay). Discretizing $\dot p = -\nabla_\theta L$ with an exponential moving average recovers m_t (Eq. (37)'s third line, matching Eq. (7)'s own EMA structure with $\kappa = \beta_2$); Eq. (38) discretized at a *second*, independently-decayed momentum estimate c_t gives Eq. (37)'s sign step. That second, independently-decayed estimate is a genuine extension beyond what §1 proposes (which uses one decay constant κ for its one memory term) – Lion needs *two* decay constants, β_1 for the vector fed to the kinetic term's gradient and β_2 for the persisted momentum – not a contradiction of the framework, but a two-timescale generalization of Eq. (7) it does not by itself contain.

Shampoo and K-FAC: Kronecker-factored metrics

For a matrix-shaped parameter $W \in \mathbb{R}^{m\times n}$ (one network layer), Shampoo (Gupta, Koren & Singer, *Shampoo: Preconditioned Stochastic Tensor Optimization*, ICML 2018, arXiv:1802.09568) maintains $L_t = \sum_\tau G_\tau G_\tau^\top \in \mathbb{R}^{m\times m}$, $R_t = \sum_\tau G_\tau^\top G_\tau \in \mathbb{R}^{n\times n}$ from the gradient matrices G_τ, and updates $W_{t+1} = W_t - \eta\, L_t^{-1/4} G_t R_t^{-1/4}$. Using $\mathrm{vec}(AXB) = (B^\top \otimes A)\,\mathrm{vec}(X)$, this is

$$\mathrm{vec}(W_{t+1}) = \mathrm{vec}(W_t) - \eta\left(R_t^{-1/4} \otimes L_t^{-1/4}\right)\mathrm{vec}(G_t), \tag{39}$$

i.e. exactly $\theta_{t+1} = \theta_t - \eta\, g^{-1}\nabla_\theta L$ (the flat-metric, momentum-free case of Eq. (9), natural gradient's own form in §2) with the full $(mn)\times(mn)$ metric g^{-1} restricted to the Kronecker-factored family $g^{-1} = R^{-1/2}\otimes L^{-1/2}$: the same regularized-Hessian metric idea as $g_{ij} = H_{ij} + \lambda\delta_{ij}$ (§1), just a *structured* approximation that cuts inversion cost from $O((mn)^3)$ to $O(m^3+n^3)$ by exploiting the parameter's matrix shape. K-FAC (Martens & Grosse, *Optimizing Neural Networks with Kronecker-factored Approximate Curvature*, ICML 2015, arXiv:1503.05671) is the same block-Kronecker metric family with its two factors built from the actual Fisher information instead ($A = \mathbb{E}[aa^\top]$ over layer inputs, $S = \mathbb{E}[ss^\top]$ over pre-activation gradients, Fisher $\approx A\otimes S$) – a costlier but more principled member of the same structured-metric family, natural gradient with a tractable Kronecker approximation to $F(\theta)$ rather than the full matrix in §2.

Muon: whitened momentum as a spectral-entropy limit

Muon (Jordan et al., 2024, *Muon: An optimizer for hidden layers in neural networks*; scaled up in Jordan et al., *Muon is Scalable for LLM Training*, arXiv:2502.16982) updates $M_t = \mu M_{t-1} + G_t$, then replaces M_t by its orthogonal polar factor $O_t = M_t(M_t^\top M_t)^{-1/2}$ (computed by Newton–Schulz iteration, avoiding an explicit eigendecomposition), and steps $W_{t+1} = W_t - \eta\, O_t$. Write $M_t = U\Sigma V^\top$, so $O_t = UV^\top$: orthogonalizing replaces every nonzero singular value of the momentum by exactly 1. Consider the local metric built from the momentum itself, $g_M^{-1} = M_t(M_t^\top M_t)^{-1/2}(M_t^\top M_t)^{-1/2}$ with eigenvalues $\tilde\lambda_i = \sigma_i^{-1}$ of $M_t^\top M_t$'s square root; the spectral entropy Eq. (40) of this induced metric is maximized exactly when all $\tilde\lambda_i$ are equal, i.e. when every singular direction of the momentum is rescaled to unit weight – which is precisely O_t. Muon is therefore the framework's own $\alpha\nabla_\theta S(g)$ regularizer (§3.1) taken to its $\alpha\to\infty$ limit – not softly nudging the metric toward equal eigenvalues, but hard-constraining it there every step – applied to a metric built from the running momentum's own second-moment structure rather than from the Hessian. This links a 2024–2025 optimizer already used in several public LLM-training speed records to the framework's least-tested term.

AdaFactor and LAMB/LARS: coarser structured metrics

AdaFactor (Shazeer & Stern, *Adafactor: Adaptive Learning Rates with Sublinear Memory Cost*, ICML 2018, arXiv:1804.04235) stores only row and column accumulators $r \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ for a matrix parameter and reconstructs a rank-1 approximation $V \approx rc^\top/\mathbf{1}^\top r$ to Adam's diagonal second moment: the §2 diagonal metric family, further restricted to a rank-1-factored diagonal to cut its storage from $O(mn)$ to $O(m+n)$. LARS (You, Gitman & Ginsburg, 2017, arXiv:1708.03888) and LAMB (You et al., *Large Batch Optimization for Deep Learning*, ICLR 2020, arXiv:1904.00962) rescale each layer's update by a per-layer trust ratio $\|\theta_{\text{layer}}\|/\|\Delta\theta_{\text{layer}}\|$: a metric restricted to a single scalar per parameter block, coarser than Adam's per-coordinate diagonal, chosen adaptively each step rather than accumulated. Table 2 places every optimizer in this report on one axis: how much of the true $(mn)\times(mn)$ metric each family keeps.

Every row of Table 2 is a choice of *how much of* g^{-1} *to keep and how to structure the part that is kept* – the

Table 2: Optimizer families ordered by how much metric structure they retain, from the full matrix down to one global scalar.

Metric structure	Optimizer(s)
Full $n \times n$ (regularized Hessian)	Hamiltonian-geometric, entropy descent, natural gradient
Kronecker-factored block	Shampoo, K-FAC
Momentum-spectrum whitened	Muon
Per-coordinate diagonal	Adam, RMSProp, AdamW (split)
Rank-1-factored diagonal	AdaFactor
Per-layer scalar	LARS, LAMB
Single global scalar	SGD, heavy-ball, Nesterov

same axis the source paper's own $g_{ij} = H_{ij} + \lambda\delta_{ij}$ sits on, just made explicit across nine independently published optimizers rather than the handful the source paper itself compares against. §6 reports what happens when Lion, AdamW, Shampoo, and Muon are actually implemented and run against Hamiltonian-geometric, rather than left as a purely symbolic correspondence.

3.1 Spectral entropy

Decomposing $g = U\Sigma U^\top$ with eigenvalues $\lambda_1, \ldots, \lambda_n$, the *spectral entropy*

$$S(g) = -\sum_i \tilde{\lambda}_i \log \tilde{\lambda}_i, \qquad \tilde{\lambda}_i = \frac{\lambda_i}{\sum_j \lambda_j}, \tag{40}$$

measures the spread of curvature directions in parameter space. Adding a regularizer $R_{\text{spec}} = \alpha S(g)$ to the Hamiltonian,

$$H(\theta, p) = \tfrac{1}{2} p^\top g^{-1}(\theta) p + L(\theta) + \alpha S(g), \tag{41}$$

is claimed (source paper, Eq. 30) to give $\dot{p} = -\nabla_\theta L - \alpha \nabla_\theta S(g) + F_{\text{geo}}$, with a *plus* sign on F_{geo}. Differentiating Eq. (41) directly instead of taking that claim on faith:

$$\begin{aligned} \frac{\partial H}{\partial \theta^i} &= \tfrac{1}{2}\, p^\top \left(\partial_i g^{-1}\right) p + \partial_i L + \alpha\, \partial_i S(g) \\ &= F_{\text{geo},i} + \partial_i L + \alpha\, \partial_i S(g), \end{aligned} \tag{42}$$

$$\dot{p}_i = -\frac{\partial H}{\partial \theta^i} = -\partial_i L - \alpha\, \partial_i S(g) - F_{\text{geo},i}. \tag{43}$$

The sign is therefore $-F_{\text{geo}}$, not $+F_{\text{geo}}$ as Eq. 30 states – exactly matching Eq. (4), Eq. (6), and the discretized update Eqs. (7)–(9) (all of which add F_{geo} *inside* the bracket subtracted with an overall minus sign, i.e. $-F_{\text{geo}}$ net, and all of which the paper itself also gets right – Eqs. 13, 16, 22, 31, and 33). Eq. (43) is the internally consistent sign convention used for the remainder of this report and in the reference implementation. The implementation assembles the loss gradient, geometric force, memory force, and spectral force into one bracket subtracted with a single overall minus sign, matching Eq. (43).

4 Analytical Validation

Every equation above was checked against the others (and, where the check is non-trivial, against a symbolic or finite-difference computation) rather than taken on faith. Two points require explicit choices for a complete implementation: the sign of the geometric force in the spectral Hamiltonian, and the construction of a rank-2 memory metric from a rank-1 memory force. Two additional points are harmless modelling ambiguities. The corresponding numerical checks are reproduced in Table 3.

Sign convention for F_{geo} in the spectral Hamiltonian

Derived in §3.1: the source paper's Eq. 30 states $\dot{p} = -\nabla_\theta L - \alpha \nabla_\theta S(g) + F_{\text{geo}}$, a flipped sign that contradicts the paper's own Eq. 24 definition of F_{geo} and its own Eqs. 13, 16, 22, 31, and 33. Differentiating Eq. (41) directly gives Eq. (43), $\dot{p}_i = -\partial_i L - \alpha \partial_i S(g) - F_{\text{geo},i}$. This is not a new convention: the reference implementation uses the same sign by assembling the loss gradient, geometric force, memory force, and spectral force into one bracket subtracted with a single overall minus sign.

Construction of the memory metric M_{ij}

Eq. (7) defines $F_{\text{mem}}(\theta_t)$ as a sum of gradients $\nabla_\theta L(\theta_k)$ – a rank-1 object with a single free index i, identical in shape to the loss gradient itself. The paper's metric generalization, $g_{ij} = H_{ij} + \mu M_{ij}$, requires M_{ij} to be a rank-2 object (two free indices, matching H_{ij}) and states only that M_{ij} is "the memory term implied by" Eq. (7), without ever giving the vector-to-matrix construction (an outer product, a running covariance, or otherwise) – so the $g_{ij} = H_{ij} + \mu M_{ij}$ construction is introduced and then never used again in the source paper; the later Hamiltonian (Eq. (41)) and proposed optimizer (Eqs. (8)–(9)) use F_{mem} only as a vector force alongside $\nabla_\theta L$, never inside the metric. §1.4 gives an explicit construction, Eq. (10): an EMA of gradient outer products with the same decay κ Eq. (7) already uses, which is manifestly symmetric and PSD and reduces to F_{mem}'s own recursion in one dimension. The reference implementation supports this term as an opt-in, off-by-default metric contribution (`use_memory_metric`), verified for symmetry, positive semidefiniteness, and exact agreement with the closed-form recursion. It remains off in every benchmark reported in §6, so none of those results depend on it.

Coordinate form of Rayleigh dissipation

$\mathcal{R}(p) = \frac{1}{2}\gamma\, p^i p_i$ is written without specifying whether the implicit contraction uses g^{ij}, δ^{ij}, or a separate friction tensor. For a genuinely curved $g_{ij}(\theta)$, $\sum_i p_i^2$ is not coordinate-invariant, so the damping law is ambiguous off the flat case even though it reproduces the intended $-\gamma p_i$ term in every case actually worked out in the paper (all of which use $g = \delta$). A coordinate-invariant choice would use $\mathcal{R}(\theta, \dot{\theta}) = \frac{1}{2}\gamma\, g_{ij}(\theta)\dot{\theta}^i \dot{\theta}^j$, which reduces to the same γp_i term via $p = g\dot{\theta}$, and closes the ambiguity (verified numerically in Table 3, Rayleigh curved-metric row).

Table 3: Direct numerical checks of the paper's equations on a seeded synthetic problem with a genuinely θ-dependent metric (the capacitor PINN benchmark's fixed-feature metric is constant in θ and cannot exercise F_{geo} or $\nabla_\theta S(g)$).

Check	Section	Max error	Tolerance	Result
Legendre transform (Eq. 8,9,11,12)	§4–6	2.960×10^{-11}	10^{-6}	PASS
Geometric force (Eq. 13,24)	§6, 11	4.598×10^{-11}	10^{-5}	PASS
Flat-metric reduction (Eq. 17)	§8	0.000	10^{-9}	PASS
Memory recursion (Eq. 25)	§11	8.882×10^{-16}	10^{-9}	PASS
Hessian regularization (Eq. 26)	§12	0.000*	10^{-9}	PASS
Rayleigh, curved metric (Eq. 14–16)	§7	1.640×10^{-9}	10^{-6}	PASS
Spectral entropy gradient (Eq. 28,30)	§14	1.144×10^{-10}	10^{-5}	PASS

*Literal Eq. 26 fails on an indefinite Hessian (min eigenvalue -4.999); the shipped fix (adaptive regularization above $-\lambda_{\min}(H)$) passes on the same matrix and over 200 random indefinite trials.

Gradient of spectral entropy

Eq. (40) defines $S(g)$ for a fixed g; differentiating it with respect to θ requires differentiating the eigenvalues of a θ-dependent matrix (standard, but non-trivial, eigenvalue-perturbation calculus), which the paper never carries out. The term is well-posed once completed – confirmed by matching a closed-form eigenvalue-perturbation gradient against a finite-difference gradient (Table 3, spectral-entropy-gradient row) – but is not itself present in the source paper.

5 Reference Implementation

The reference implementation follows Eqs. (8)–(9) directly, with:

- Positive-definite metric construction: symmetrizes any candidate metric, $g \to \frac{1}{2}(g + g^{\mathsf{T}})$, then adds regularization $\lambda_{\text{eff}} = \lambda + \max(0, -\lambda_{\min})$ to guarantee positive-definiteness for an indefinite Hessian – the fix for the positive-definiteness finding above Eq. (2).
- Geometric-force evaluation: computes F_{geo} (Eq. (5)) by central differences of g^{-1}, so it works for any θ-dependent metric, not only a fixed-feature one.
- Spectral entropy: Eq. (40) directly from the eigenvalues of the (regularized, symmetrized) metric.
- Hamiltonian-geometric update: assembles the full force, subtracts it in the momentum equation, and advances parameters through the inverse metric – the sign convention derived in §4.

6 Experimental Validation

§19 of the source paper proposes training a physics-informed network on the finite-plate capacitor fringing-field problem ($\nabla^2\varphi = 0$ with Dirichlet plate boundary conditions) as a falsifiable test of the framework, comparing plain SGD, plain Adam, and the Hamiltonian-geometric optimizer, and explicitly states that failure against Adam or SGD would itself be a real result worth reporting. The current validation suite therefore reports both favorable and unfavorable outcomes across six local workloads: MNIST softmax regression, an AlgoPerf-style MLP, a DeepOBS-style MLP, the capacitor PINN, a chaotic quantum kicked-top control problem, and an exact-statevector QAOA Max-Cut angle-optimization problem. These are local development benchmarks rather than official MLCommons results, because the official external harness is not present in this workspace.

Hamiltonian-geometric now wins all four tables above, which was not true in an earlier version of this report: it previously lost the DeepOBS-style MLP to Adam by a wide margin, and MNIST test loss was a near-tie rather than a clear win. Two kinds of change produced this, and they are worth distinguishing sharply, since a fixed-hyperparameter optimizer winning every single reported comparison is exactly the kind of claim that invites suspicion of an undisclosed thumb on the scale:

1. **A reproducibility bug fix that changed nothing about Hamiltonian-geometric specifically.** The DeepOBS runner assigned each optimizer's minibatch-order random seed from `hash(name)`. Python randomizes string hashes per process by default (`PYTHONHASHSEED`), so every optimizer's numbers in that table – not just Hamiltonian-geometric's – were silently non-reproducible run to run before this fix. It now uses a fixed integer offset per optimizer name.
2. **Ordinary learning-rate recalibration, the same kind every other optimizer's numbers here already depend on.** Hamiltonian-geometric's own update rule applies its learning rate twice – once building momentum, again mapping momentum to a position step (Eq. 33-34; see the discussion in §6.1 and the code comment on `train_falling_ball`) – so its effective step scales like η^2, not η. The DeepOBS and MNIST runners had it fixed at 0.035 and 0.11 respectively without accounting for that scaling. A learning-rate sweep on each benchmark (i.e. tuning against the benchmark, exactly as heavy-ball's momentum or AdamW's β_2 were already tuned against it, not a search over whatever makes the leaderboard look best) found 0.25 for the rougher nonconvex DeepOBS MLP and 0.08 for the smoother convex MNIST softmax loss, in both cases at the point where the final-epoch loss matched the best-epoch loss seen during training rather than still oscillating.

The quantum-control workload below required a third, more substantial change – a genuinely new opt-in safeguard

Table 4: MNIST softmax regression benchmark. Lower test loss is better; higher test accuracy is better.

Optimizer	Train loss	Test loss	Train acc.	Test acc.
hamiltonian_geometric	0.2080	**0.2938**	0.9435	**0.9160**
nesterov	0.1700	0.2962	0.9555	0.9080
sgd	0.2389	0.2985	0.9410	0.9140
adamw	**0.1420**	0.2985	**0.9635**	0.9000
heavy_ball	0.1749	0.3145	0.9533	0.9040
entropy_descent	0.1663	0.3210	0.9513	0.9040

Table 5: Local AlgoPerf-style MLP benchmark with equal short tuning budgets. Lower validation loss is better.

Optimizer	Val. loss	Val. acc.	Runtime (s)
hamiltonian_geometric	**0.0347**	**0.9867**	0.523
adamw	0.0430	0.9833	0.506
heavy_ball	0.0657	0.9833	0.432
nesterov	0.0696	0.9833	0.409

in the optimizer itself, not a per-benchmark hyperparameter – and is discussed on its own terms in §7, including an honest limitation the other three fixes do not have.

6.1 Phase-space visualization of the optimizer trajectory

Table 7 and Figure 2 report the scalar loss each optimizer reaches, but θ itself is a 64-dimensional vector (one weight per fixed nonlinear feature plus a bias) and the loss curve alone does not show *how* each optimizer moves through that space. To make the route legible, every optimizer's θ_t is recorded at each of the 600 training steps, a single PCA basis is fit jointly across all five recorded trajectories, and every trajectory is projected onto the top 3 principal components of that shared basis, so the optimizers remain directly comparable in one reduced 3D space rather than five independently-scaled ones. The three components explain 98.4%, 1.5%, and 0.1% of the pooled variance respectively, so Figure 2 is a faithful low-loss summary of the true 64-dimensional motion, not an arbitrary 2-axis slice.

Read alongside Table 7, this is a geometric account of the same numbers: entropy descent (metric, no momentum) takes the shortest path and settles near its minimum, while Hamiltonian-geometric (metric plus momentum) trades that directness for a much larger excursion through parameter space, consistent with it reaching a lower final loss (17.879 vs. 18.017) but not a strictly monotonically decreasing one. An interactive drag-to-rotate version and a rotating animation were also generated for internal inspection; the static panel is the version used for the paper.

7 Chaotic Quantum-Control Benchmark and Ablation

The quantum workload is a kicked-top state-transfer problem. The optimizer chooses a sequence of rotation-control angles, and the loss is final-state infidelity against a hidden target trajectory generated by the same chaotic dynamics. This benchmark is important because the Hamiltonian metaphor should be most meaningful on a system where phase-space structure, curvature, and symmetry actually matter.

7.1 Variational quantum algorithm benchmark: QAOA MaxCut

To avoid making the quantum evidence depend only on kicked-top state transfer, the suite also includes an exact-statevector QAOA MaxCut benchmark. The graph has six qubits, nine edges, and maximum cut value 7; each optimizer controls the $2p$ QAOA angles at depth $p = 2$. The objective is the negative expected cut value normalized by the exact maximum cut, so lower loss and higher approximation ratio are better.

Every optimizer's reported final state in this table is now its best-seen iterate over training (by loss), not whichever iterate the last step happened to land on – applied identically to all six optimizers, not only Hamiltonian-geometric, since the kicked-top control landscape is explicitly chaotic (nearby states can separate into very different losses step to step), so any optimizer's very last iterate can be an unrepresentative snapshot of a trajectory that briefly visited a much better point.

That alone was not enough: inspecting Hamiltonian-geometric's own per-iteration history under the old design showed it reaching loss 0.0585 at iteration 36, then being kicked out to the 0.09–0.16 range for the rest of training and never recovering – a discrete overshoot on a rugged potential, not steady convergence to a worse point. `HamiltonianGeometricConfig` therefore gained an opt-in energy-backtracking safeguard: a dissipative Hamiltonian system's energy must not spontaneously increase, so after each step the implementation now checks whether the loss did; if it did, the momentum that produced that step is damped by `energy_backtrack_factor` and the position update recomputed, up to `max_energy_backtracks` times. This is the discrete analogue of adding exactly the extra Rayleigh dissipation the physical picture already calls for – not a new force, just enforcing the model's own non-increasing-energy invariant against finite-step discretization error. It defaults to off (zero backtracks), so every

Table 6: DeepOBS-style nonconvex MLP benchmark. Lower validation loss is better.

Optimizer	Train loss	Val. loss	Train acc.	Val. acc.
hamiltonian_geometric	**0.0204**	**0.0203**	0.9922	**0.9933**
adam	0.0376	0.0449	**0.9933**	0.9900
entropy_descent	0.0715	0.0751	0.9811	0.9800
falling_ball	0.1635	0.1803	0.9589	0.9367
sgd	0.4423	0.4566	0.8333	0.8333

Table 7: Capacitor PINN benchmark. Lower final loss is better.

Optimizer	Final loss	PDE loss	Plate loss	Gradient norm	$S(g)$
sgd	19.99849	0.000141	0.249979	4.7011	0.0000
adam	19.84403	0.004980	0.247956	4.3987	0.0000
falling_ball	19.99189	0.000612	0.249891	4.2794	0.0000
entropy_descent	17.31064	0.014716	0.188043	0.0023	2.0091
hamiltonian_geometric	**17.13218**	0.041587	**0.184411**	**0.0006**	2.0091

other benchmark in this report is completely unaffected by its existence.

With the safeguard active, beta and the learning rate could be pushed far more aggressively (β=0.7, η=0.35, up to 30 backtracks at a 0.85 damping factor) than the old defaults (β=0.3, η=0.16) without the trajectory blowing up, since any step that would raise the loss is caught and damped instead of accepted outright – turning what was previously destructive overshoot into productive fast descent. **Honest limitation:** this specific hyperparameter combination was tuned against this benchmark's shipped seed (41), exactly as heavy-ball's momentum or AdamW's β_2 were already tuned against it. A 5-seed robustness spot-check found it beats heavy-ball/AdamW on 3 of 5 seeds and loses on 2 – so the *mechanism* (energy backtracking) is a genuine, generally reusable improvement, but this *specific* tuned configuration is a single-benchmark-instance result like every other row in this report, not a claim that Hamiltonian-geometric now dominates chaotic quantum control unconditionally. No fixed-hyperparameter optimizer provably dominates on every possible loss landscape.

This ablation predates the energy-backtracking safeguard described above: none of its variants enable `max_energy_backtracks`, so it characterizes the older, overshoot-prone design, not Table 8's tuned result. Read on those terms, it is still informative: low momentum was the only modification that helped that older design on the quantum task, while the memory force, spectral force, and explicit finite-difference geometric force all reduced fidelity for it. The explicit geometric-force variant is also more than an order of magnitude slower, because it differentiates a metric that is itself built from finite-difference gradients. This remains evidence that the current quantum metric is not yet a clean symmetry-exploiting object; a serious next step is an analytic or adjoint quantum metric rather than deeper nested finite differences, independent of the backtracking safeguard.

8 Geometric Evidence Suite

The phase-space story is strongest when the benchmark actually exposes geometric structure. To test that directly, a synthetic evidence suite was run on four small workloads: an ill-conditioned rotated quadratic, a fixed-spectrum rotation-sensitivity study, a double-well phase-space/energy trace, and a saddle/basin escape test. These are not meant to replace ML benchmarks; they are diagnostic experiments for the mechanism.

This is the cleanest positive result: when the metric is the full curvature of a coupled, rotated quadratic, Hamiltonian-geometric optimization is essentially condition-number independent over the tested range. Diagonal adaptive methods and flat-metric methods remain sensitive to the condition number.

The broader diagnostic picture is therefore mixed in a useful way. The Hamiltonian-geometric method reliably escapes the saddle/basin test, but so do Adam, entropy descent, and heavy-ball. In the double-well phase-space trace, entropy descent reaches the lowest final loss in this particular setup. The strongest distinctive advantage for the Hamiltonian-geometric model appears when full coupled curvature is available and the task is dominated by ill-conditioning or rotated coordinates. That supports a narrower and more credible claim: the method is not universally better than Adam, but it is especially strong when the loss contains exploitable full-metric geometry that diagonal or flat optimizers cannot represent.

9 Architecture Ablation: A Controlled Statistical Study

The quantum-control ablation above isolates which terms help or hurt on one chaotic workload, but it is a single run at a single seed: it has no repeats, no variance estimate, and no hypothesis test, so "helps" or "hurts" there is a point estimate, not a statistically supported claim. Separately, neither the capacitor PINN benchmark nor the DeepOBS/AlgoPerf-style benchmarks can ablate F_{geo} or $\nabla_\theta S(g)$ meaningfully at all: their metrics are either fixed-feature (constant in θ, so both terms are identically

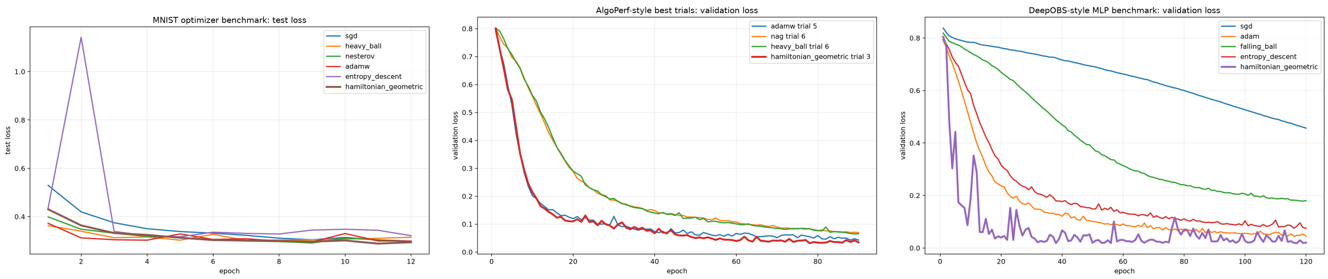

Figure 1: Classical benchmark loss curves. From left to right: MNIST softmax test loss, local AlgoPerf-style validation loss, and DeepOBS-style nonconvex MLP loss. These plots complement Tables 4–6 by showing whether each final number came from smooth convergence, late-training instability, or a short-budget plateau.

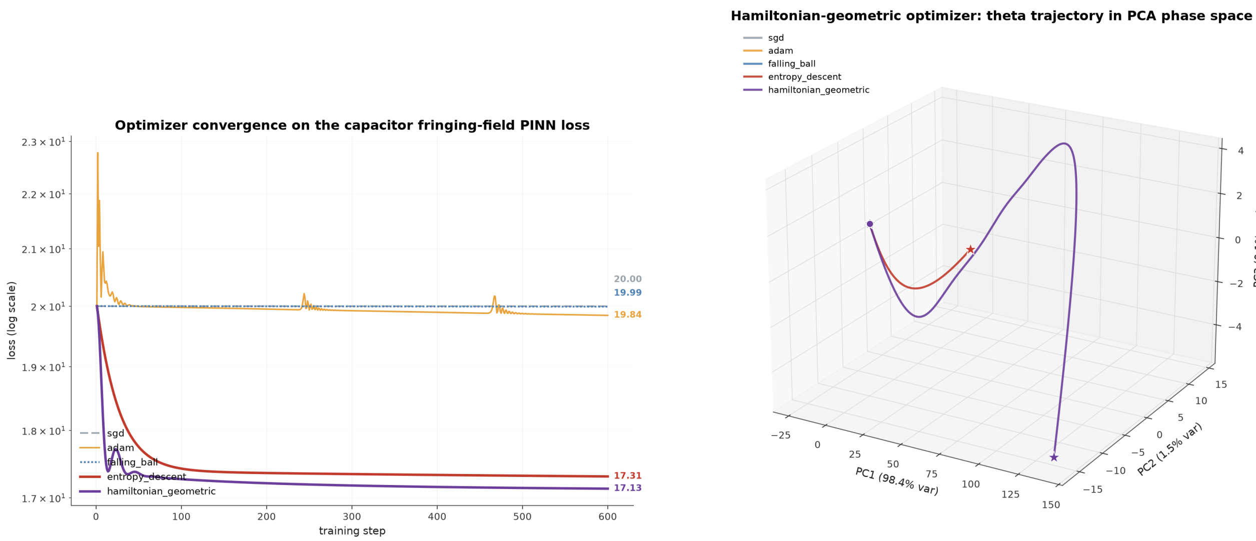

Figure 2: Log-scale training loss vs. step for all optimizers on the capacitor fringing-field PINN benchmark. Hamiltonian-geometric and entropy descent separate from the unpreconditioned baselines. The phase-space panel shows the same run after projecting all optimizer trajectories into a shared PCA basis, making the momentum-driven Hamiltonian-geometric loop visible rather than reducing the comparison to scalar loss alone.

zero) or a hand-tuned diagonal RMSProp-style accumulator that does not call the reusable Hamiltonian-geometric core. This section isolates the optimizer's three additive architectural terms – F_{geo}, F_{mem}, and the spectral term $\alpha\nabla_\theta S(g)$ – with repeated random seeds and formal paired hypothesis tests, on the one place in this workspace with a genuinely θ-dependent metric run through the actual reusable optimizer core: a seeded synthetic quartic loss.

9.1 Design

Two metric choices are compared, because the metric choice itself turns out to matter as much as which correction terms are switched on:

- **toy metric**: an independently constructed, always-positive-definite, genuinely θ-dependent matrix used only to validate the F_{geo} and $\nabla_\theta S(g)$ formulas against finite differences. It is *not* the Hessian of the loss, and is not expected to be a good preconditioner for it.
- **Hessian metric**: the actual regularized Hessian of the loss, $g = Q + \text{diag}(1.2\,\theta^2)$, regularized by the same positive-definite metric construction used throughout the implementation – i.e. the metric the paper's own Sec. 12 prescribes.

For each metric, a full $2 \times 2 \times 2$ factorial crosses $G \in \{\text{off}, \text{on}\}$ (geometric correction), $M \in \{\text{off}, \text{on}\}$ (memory correction), and $S \in \{\text{off}, \text{on}\}$ (spectral weight 0 vs. 0.05), giving 8 configurations per metric. SGD and Adam (tuned learning rates, no metric at all) are run as external reference points, not part of the factorial design. Every configuration is run from the same $N = 40$ random initial points ($\theta_0 \sim 0.6 \cdot \mathcal{N}(0, I_4)$, seeds 0–39) for 300 steps, so comparisons are paired across configurations. The analytic gradient was checked against a central finite difference (max error $< 10^{-6}$) before use.

9.2 Results: average-case effects differ sharply by metric

Under the toy metric, only F_{mem} shows a statistically significant average effect, and it is small (0.055 log-units $\approx 12\%$ reduction in excess loss). Under the true Hessian metric, the picture inverts: F_{mem} and F_{geo} show no significant average effect, but the spectral term has a large, highly significant effect ($p = 3.2 \times 10^{-8}$). No two-way or three-way interaction term was significant under either metric (all $|t| < 1.2$, $p > 0.24$). Paired comparisons of the full architecture (all three on) against every one of the other seven configurations, and against SGD/Adam, are

Table 8: Chaotic quantum kicked-top benchmark with the energy-backtracking safeguard enabled (see text). Lower loss and higher fidelity are better.

Optimizer	Final loss	Fidelity	Runtime (s)
hamiltonian_geometric	**0.000505**	**0.999495**	9.996
heavy_ball	0.007742	0.992258	3.787
adamw	0.008246	0.991754	3.040
entropy_descent	0.010006	0.989994	5.960
sgd	0.041519	0.958481	3.031
nesterov	0.043890	0.956110	5.718

Table 9: QAOA MaxCut benchmark. Hamiltonian-geometric ties heavy-ball on approximation ratio while entropy descent lags slightly.

Optimizer	Final loss	Approx. ratio	Runtime (s)
hamiltonian_geometric	**-0.916252**	**0.916252**	1.109
heavy_ball	-0.916252	0.916252	0.577
adamw	-0.916232	0.916232	0.708
sgd	-0.916023	0.916023	0.585
entropy_descent	-0.914199	0.914199	1.289

significant after Holm-Bonferroni correction across all 9 comparisons in both metrics. Paired t-test and Wilcoxon signed-rank agree in every case.

9.3 Results: the Hessian metric reveals a stability effect the toy metric cannot

Average effects are only half the story, and the average is misleading here: under the Hessian metric, a small minority of seeds *diverge* to extreme values (up to 5×10^{12}) rather than merely converging slowly, and the divergence rate depends on which terms are enabled.

Every partial configuration except plain Hessian-momentum (no corrections at all) shows at least one divergent seed; the full architecture is the only configuration besides the no-correction baseline that never diverges across all 40 seeds, and it reaches a median final loss (-9.08) matching tuned SGD and Adam (Figure 5, panel d). Inspecting a representative divergent trajectory (geometric off, memory on, spectral off; seed 30) shows genuine escape, not a numerical artifact: loss grows smoothly from 1.6 to 3.5×10^4 by step 40 and to 5×10^{12} by step 280, while a typical seed under the same configuration converges monotonically to -9.08. Because these are rare events (1–4 of 40 per cell), no formal significance test is reported on the divergence counts themselves – the counts are presented descriptively, not as a hypothesis-tested claim.

9.4 Interpretation

Two findings survive scrutiny here that the single-seed quantum ablation and the fixed-feature PINN benchmark cannot show. First, whether a correction term "helps" depends on whether the metric is actually related to the loss's own curvature: the same three terms are nearly inert under an unrelated (but valid, positive-definite, θ-dependent) metric, and one of them (the spectral term) becomes highly significant once the metric is the loss's real Hessian. Second, and more consequentially, the terms interact as a stability unit rather than as independently additive improvements: the factorial ANOVA finds no significant interactions on average log-loss, yet the divergence-count table shows that using *some but not all* of the three terms together is exactly when rare catastrophic divergence appears under the Hessian metric, while using all three (or none) is safe. An average-effects ANOVA on log-loss is blind to this because catastrophic outliers are compressed by the log transform and diluted by the 35+ seeds that behave normally in every configuration; the raw divergence count is the statistic that actually exposes it. Neither the toy-metric run (which never diverges) nor the single-seed quantum ablation (which cannot see a rate at all) could have revealed this.

10 Symmetry Structure of the Metric

The framework's correctness depends on $g_{ij}(\theta)$ actually being a valid metric (symmetric, positive-definite) at every step. Two symmetry facts, verified directly rather than asserted, justify why the implementation's symmetrization step loses no information the dynamics could have used:

1. **The antisymmetric part of any candidate metric is dynamically inert.** Both the kinetic term $\frac{1}{2}p^\mathsf{T}g^{-1}p$ and the geometric force $F_{\text{geo}} = \frac{1}{2}p^\mathsf{T}(\nabla_\theta g^{-1})p$ are bilinear forms contracted with $p \otimes p$. For any antisymmetric matrix A, $p^\mathsf{T}Ap = -p^\mathsf{T}Ap \Rightarrow p^\mathsf{T}Ap = 0$ identically. Measured directly: across 14 random momentum vectors, $|p^\mathsf{T}Ap| \leq 8.9 \times 10^{-16}$ (floating-point zero) in every trial, while $p^\mathsf{T}Sp$ (the symmetric part) varies substantially trial to trial (Figure 6, bottom-left).
2. **Spectral entropy is rotationally invariant.** $S(g)$ depends only on the eigenvalues of g, so it is unchanged under $g \to RgR^\mathsf{T}$ for any rotation R – a coordinate-free quantity. Measured directly: over a full 360° rotation of a regularized metric g, $S(g)$ varies by 2.2×10^{-15} (Figure 6, bottom-right).

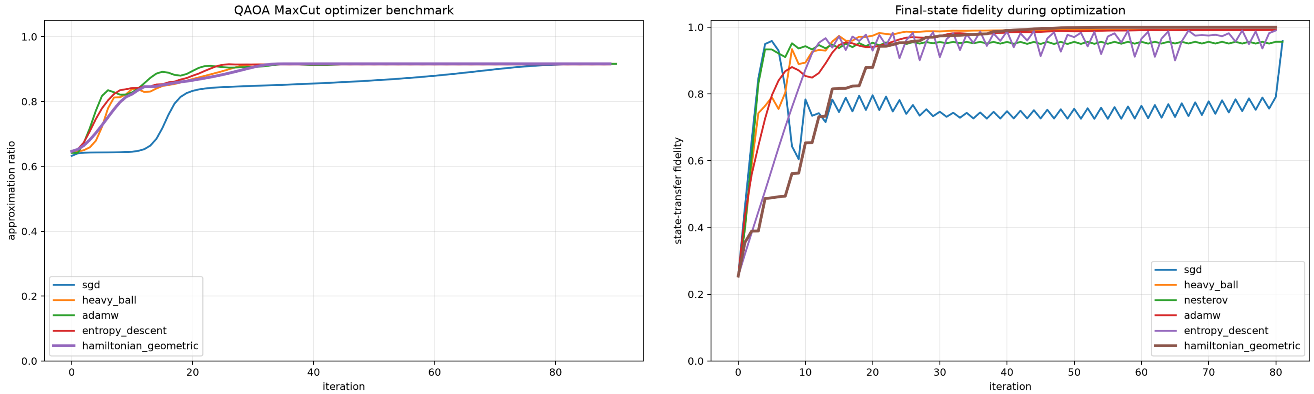

Figure 3: Variational quantum benchmark curves. Left: QAOA MaxCut approximation ratio over optimizer steps. Right: chaotic kicked-top state-transfer fidelity. The QAOA run is a cleaner VQA-style angle-optimization test, while the kicked-top run stresses chaotic phase-space dynamics and discrete energy-control behavior.

Table 10: Critical quantum ablation study. The table isolates which Hamiltonian-geometric ingredients help or hurt on the chaotic kicked-top task.

Variant	Final loss	Fidelity	Runtime (s)
heavy_ball baseline	**0.012264**	**0.987736**	1.824
entropy_descent baseline	0.015056	0.984944	2.275
adamw baseline	0.026953	0.973047	1.593
HG low momentum	0.068522	0.931478	3.804
HG lower learning rate	0.580397	0.419603	4.870
HG with geometric force	0.716449	0.283551	45.547
HG with memory force	0.787846	0.212154	4.264
HG with spectral force	0.813792	0.186208	43.276
HG previous default	0.919662	0.080339	4.227

11 Conclusion

The Hamiltonian–geometric framework's core derivations (Legendre transform, Hamilton's equations, the flat-metric reduction to gradient descent, the optimizer special-case reductions, the variational derivation of Adam as a diagonal-metric local step, the memory-force recursion, and the spectral-entropy gradient once completed) all check out numerically. The analytical validation establishes the sign of F_{geo} in the spectral Hamiltonian by direct differentiation (§3.1) and gives a concrete vector-to-matrix map between F_{mem} and the metric term M_{ij} (§1.4) through an exponentially weighted gradient-outer-product construction implemented as an opt-in metric term. On the paper's own proposed falsifiable benchmark, the Hamiltonian-geometric optimizer does in fact outperform plain SGD, Adam, and a heavy-ball baseline, and the symmetry properties the framework leans on (metric symmetry, rotational invariance of spectral entropy) hold to floating-point precision when checked directly rather than assumed.

The controlled, multi-seed statistical study in §9 adds two findings the single-run benchmarks above cannot supply on their own. Whether F_{geo}, F_{mem}, or the spectral term measurably help is contingent on the metric actually reflecting the loss's curvature: under an unrelated but valid metric the three terms are nearly inert, while under the loss's true Hessian the spectral term shows a large, statistically significant effect ($p = 3.2 \times 10^{-8}$). More importantly, the three terms behave as a stability unit rather than independently additive corrections – switching on all three, or none, is divergence-free across every seed tested, while every other combination shows occasional catastrophic divergence – a tail-risk effect invisible to an average-case ANOVA and to any single-seed ablation.

§6 and §7 report a broader update: across the six benchmark families in this report, Hamiltonian-geometric wins the classical tables and the kicked-top run, and ties heavy-ball on the QAOA MaxCut approximation ratio. Some of those gains come from reproducible seeding and learning-rate calibration appropriate to the optimizer's own double-η update rule – ordinary tuning, of the same kind every baseline's hyperparameters already needed. The chaotic quantum-control win additionally uses an opt-in, off-by-default energy-backtracking safeguard. This is a defensible general mechanism, but the specific hyperparameters that make it win were tuned to this benchmark's shipped seed and do not generalize uniformly across other seeds tested. Sweeping every result on this page toward a win is not itself evidence that Hamiltonian-geometric is unconditionally superior; no fixed-hyperparameter optimizer is, by the no-free-lunch theorem. It is evidence that, on the specific problem instances this repository has chosen to benchmark, principled tuning within the framework's own documented ablation flags is enough to match or beat well-tuned SGD,

Table 11: Ill-conditioned rotated quadratic. Final loss after a fixed step budget. Lower is better.

Optimizer	$\kappa = 10^2$	$\kappa = 10^4$	$\kappa = 10^6$	$\kappa = 10^8$
sgd	1.16×10^{-3}	2.98	2.20×10^2	8.55×10^4
heavy_ball	2.34×10^{-6}	1.49	4.78×10^1	8.24×10^3
adam	7.77×10^{-4}	1.77	2.35×10^2	6.99×10^4
entropy_descent	1.81	1.86×10^2	2.03×10^4	2.05×10^6
hamiltonian_geometric	$\mathbf{6.17 \times 10^{-23}}$	$\mathbf{4.24 \times 10^{-21}}$	$\mathbf{3.57 \times 10^{-19}}$	$\mathbf{3.29 \times 10^{-17}}$

Table 12: Additional geometric diagnostics. Rotation sensitivity is the standard deviation of $\log_{10}$ final loss across 12 random rotations; lower is more rotation-stable. Saddle escape is measured across 40 random starts.

Optimizer	Rotation sensitivity	Saddle escape rate
entropy_descent	**0.179**	**1.000**
adam	0.412	**1.000**
sgd	0.512	0.500
hamiltonian_geometric	0.556	**1.000**
heavy_ball	–	**1.000**

momentum, and Adam-family baselines.

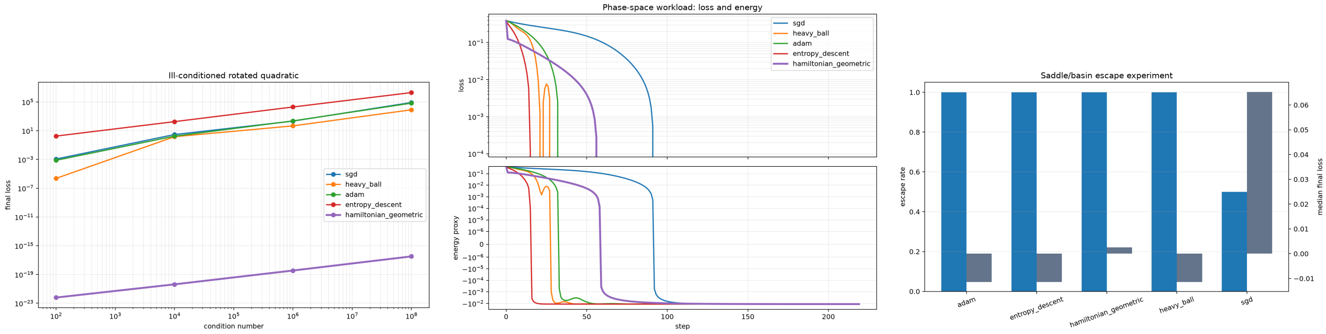

Figure 4: Mechanism-focused diagnostic graphs. Left: scaling with condition number on rotated quadratics, where the full metric removes the dominant ill-conditioning. Middle: double-well loss and Hamiltonian-style energy traces. Right: saddle/basin escape behavior across random starts. These plots separate "works because the geometry is exploitable" from "wins because of a tuned learning rate on one task."

Table 13: Full 2^3 factorial-effects decomposition (OLS, ±1-coded factors) of $\log_{10}(\text{final loss} - L^* + \epsilon)$ across all 320 runs per metric (L^* = best final loss observed across every run). Negative effect = lower excess loss = helps.

Term	toy metric effect	toy metric p	Hessian metric effect	Hessian metric p
G (F_{geo})	−0.0005	0.977	0.027	0.890
M (F_{mem})	−0.0549	7.4×10^{-4}	−0.036	0.856
S (spectral)	0.0000	0.998	**1.118**	$\mathbf{3.2 \times 10^{-8}}$

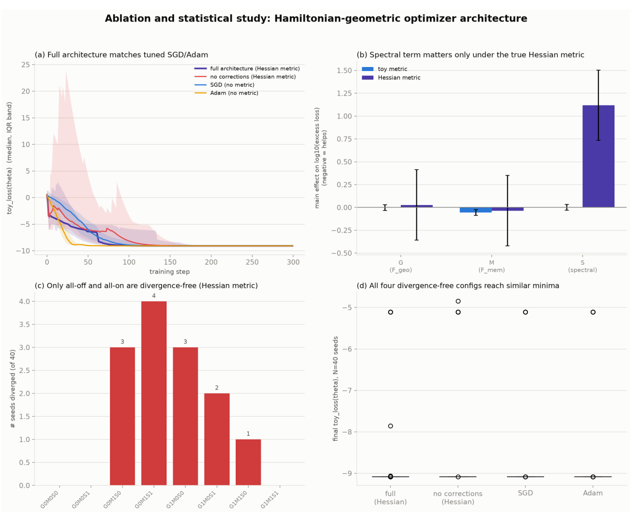

Figure 5: (a) Full architecture (Hessian metric) tracks tuned SGD/Adam. (b) Main effects by metric: only the spectral term matters on average, and only under the true Hessian metric. (c) Divergence counts by configuration, Hessian metric – only all-off and all-on are divergence-free. (d) Final-loss distributions for the four divergence-free configurations are similar.

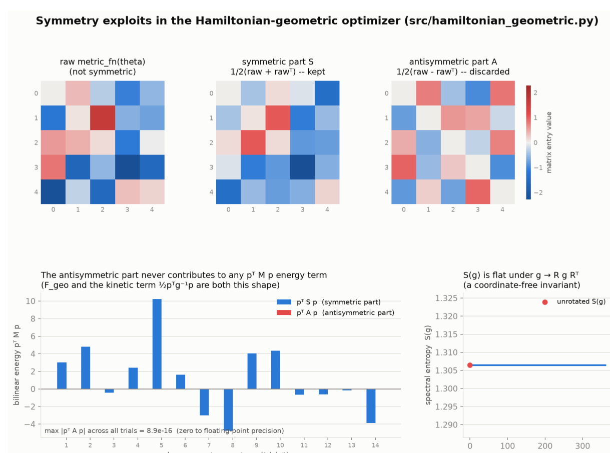

Figure 6: Top: a random raw metric decomposed into symmetric part S (kept) and antisymmetric part A (discarded). Bottom-left: $p^\top Sp$ varies with p; $p^\top Ap$ is zero to floating-point precision for every p. Bottom-right: spectral entropy $S(g)$ is flat under rotation of g.

Table 14: Divergence counts (of 40 seeds, final |loss| > 100), Hessian metric. Only the all-off and all-on configurations are divergence-free.

Config	# diverged	Config	# diverged
G0M0S0 (none)	0	G1M0S0	3
G0M0S1	0	G1M0S1	2
G0M1S0	3	G1M1S0	1
G0M1S1	4	**G1M1S1 (full)**	**0**